

AI Didn't Replace PowerShell

It Made It More Important

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Agenda

Opening — PowerShell meets the agent era

Skills architecture — discover, install, create

PowerShell skills — Pester · Azure Automation · Sampler

MCP servers — AI calling PowerShell

Balance — agent boss vs. code writer

Wrap-up — one skill, one MCP server

The shift

AI did not make PowerShell less important.
It changed where PowerShell expertise shows up.

Yesterday's job

Turn intent into reliable automation. Write the script. Own the result.

Today's job

Teach agents what good PowerShell looks like. Give them the right tools. Review what they produce. Decide when to let them keep going — and when to drop back into the code.

The "agent boss" role — setting direction, providing context, choosing the right specialist, checking the result, keeping accountability.

If you tried AI once and walked away...

"There's something I call the Siri phenomena. I used Siri 15 years ago and decided it sucked. And if you ask me today, I'll tell you it sucks — even though I haven't used it in 15 years."

— Jeffrey Snover, *The Guy Who Invented PowerShell Is Blown Away By AI*

Trying AI once when it was bad and dismissing it forever — while the field moves at lightning speed.

Hit F5 on your thinking

"There are plenty of things to remain skeptical about AI — but if you're still in the camp that thinks AI is just a parlor trick, you need to hit F5 on that thinking."

— Jeffrey Snover, LinkedIn

"Your knowledge is your enemy. What used to be unreasonable is now reasonable. What used to be impossible is now possible."

— Jeffrey Snover

PowerShell was never just a tool

It was always a mindset

Verb-Noun

Predictable surface area. The discoverability that lets a stranger — human or agent — guess your next command.

Object pipeline

Structured data flowing between commands. Not text to re-parse. A framework for understanding systems.

Predict-before-Enter

Read the command. Reason about what it will do. Decide whether to run it. The exact discipline an agent boss needs.

Community lineage: Perl had CPAN. PowerShell has the Gallery. We've shared executable knowledge for decades.

"The mouse is intrinsically antisocial. You can't share how you moved it. Scripts are shareable. That's why CLI communities form."

— Jeffrey Snover, *Monad to Millions*

Skills Architecture

Package PowerShell expertise so an agent can load it on demand — instead of dragging every instruction into every conversation.





Demo • Skill marketplace

[demos/skills-architecture/marketplace/](#)

Discover → Install → Use

Skills are knowledge an agent can load on demand. Today you don't have to write the first one.



"The agent didn't get smarter. It got *specifically* smarter."



Demo · Custom skill creator

<demos/skills-architecture/custom-skill-creator/>

Author your own

The skill is the conversation you wish you didn't have to repeat. Capture it once. Hand it to every future agent — and every future teammate.



"The agent does the writing. You supply the knowledge. Free for everyone who comes after — including the agent."

SECTION 2 OF 3

PowerShell Skills

From "skills exist" to "skills that capture the PowerShell craft." Three skills. Three jobs you already do — done better by the agent because *you* wrote down what good looks like.



Demo • Sampler module skill

`demos/skill-demos/sampler-module/`

Sampler is opinionated. That's the point. The skill makes "scaffold a new module the way our community ships modules" a single prompt.



"From idea to a shippable module shape in one prompt — because the conventions are in the skill, not your head."



Take-home • Pester testing skill

`demos/skill-demos/pester-testing/`

Pester has 15 years of community conventions. The skill teaches the agent *your team's* flavour of them.



"Without the skill it writes Pester. With the skill it writes *your* Pester."



Take-home · Azure Automation skill

[demos/skill-demos/azure-automation/](#)

[show & tell · run it from the repo](#)

Runbooks have a thousand sharp edges — managed identity, hybrid worker, output streams, error handling. The skill knows them all so you don't have to re-explain them every prompt.



"Managed identity by default. Because the skill said so."

SECTION 3 OF 3

MCP Servers

Skills give agents *knowledge*.
MCP gives agents *reach*.



MCP — the universal connector

Open protocol · USB-C for agents · bring your own tooling

What it is

- Open protocol for AI ↔ tool calls
- **JSON-RPC over stdio or HTTP** — simple wire format
- Standardized tool descriptions & invocation
- Client-neutral — VS Code, Claude Code, Copilot CLI
- Server-neutral — PowerShell, Python, Node, Go...

Why we care

- Expose PowerShell as a live execution surface
- Agent discovers what your code can do
- No bespoke SDK, no deep AI internals
- Reuse existing scripts, modules, runbooks

Agent (VS Code / CLI) —MCP→ PowerShell MCP server → Your code / Azure / on-prem
(tool descriptions, invocations, results)

Same config works in VS Code Copilot agent mode, Claude Code, and the Copilot CLI. The server doesn't care which client. The client doesn't care which server.

The super-simple PowerShell MCP server

Inspired by Doug Finke — how shallow this actually is



"Use MCPs to read your existing infrastructure. Ask AI to review it. Map a migration. Don't boil the ocean — break it into chunks AI can reason over."

— Doug Finke

Param block + comment-based help is already 80% of an MCP tool description. **ValidateSet** becomes the input schema. **The LLM decides WHEN to call. You decide WHAT is callable.**



Demo · Local PowerShell MCP — InfraOps

demos/mcpserverps-infraops/ (local)

The MCP server is just a script. Same script works against a local model if your data can't leave the building.



"It's a script. The protocol doesn't care if the model is in Redmond or in your basement."



Demo • InfraOps on Azure

`demos/mcpserverps-infraops/ (Azure)`

Same four tools, hosted on Azure Container Apps. Two networking paths from one MCP surface: direct VNet SSH to demo VMs, Arc HybridConnectivity tunnel to Arc-enabled servers.

"Demo-grade auth today. Production-grade auth is the next conversation — but the shape doesn't change."

It's not about the LLMs

"We used to say to PowerShell: 'do my job and order me a pizza.' Now we do it with AI. And read my mind. But — it's not about the LLMs. It's about **the harness**."

— Doug Finke, closing of *Monad to Millions*

Steinberger put an agent in a minimal container — no curl, no tools. The agent said: "*There's literally nothing here. I'm sad lobster now.*" Then it wrote Lobster Curl 0.1 in C, found a compiler, opened a socket, and started reading websites. "*I have a big lobster. I built my own curl.*"

— Peter Steinberger, on Hanselminutes

Give agents the tools you want them to use — **before they build their own.**

*"Buckle your seatbelt,
Dorothy.
'Cause Kansas is going bye-
bye."*

Balance

Skills and MCP make agents more capable.
They don't tell you *when* to use them.

"Should the agent do it, or should I?"

The "just specify it fully" trap

"If you fully specified it, you would have used keywords in a programming language — and then you would control the thing yourself."

— Mark Russinovich, with Scott Hanselman · *Learn to Have Taste*

Full specification **is** programming. The only difference is syntax.

Which is exactly why PowerShell still matters. It's the specification language we already speak — verb-noun, typed parameters, structured output. When the spec needs to be precise, you reach for code. That's not a regression. That's the job.

The new split

Adam Jacob — AI Native Infrastructure Automation



YOUR JOB

Design · Plan

Architecture. Vocabulary. Intent. Constraints. The conversation with yourself about what the code is supposed to do.

AGENT-ASSISTED

Build · Test · Operate

Implementation, generated tests, PR review against your standards, remediation runs.

At 50,000 lines of working code a day, you can't read the code. Design and planning become the critical human work.

Trade-offs, named

"Our entire industry is built on crappy, unreliable systems. Ethernet — crappy and unreliable. So you put TCP/IP on top of it. AI is crappy and unreliable. Where it matters, develop the strategy for detecting the errors and compensating. Code is what you do with code."

— Jeffrey Snover, *Monad to Millions*

Velocity ↑, Understanding ↓

Delegating more means knowing less, faster. Fine for throwaway code. Catastrophic at 3am with no muscle memory.

Coverage ↑, Depth ↓

Skills let one person cover more domains — but the depth comes from the people who *wrote* the skills. If everyone delegates and nobody writes, the system rots.

"You need more sleeps. Taste can't be bootcamped."

— Scott Hanselman

No rule fixes this. There's just the discipline of asking — every time.

What the agent boss does every week

1 • One skill

Add or update one SKILL.md that captures domain knowledge. 20 minutes. Free for everyone after.

2 • One MCP tool

Connect or audit one MCP tool that gives agents real-world reach. Schemas are the boundary.

3 • One review

Review one piece of agent-generated work against the skill / tool that *should* have guided it.

Not "use AI more." Not "use AI less." Just: be the person who shapes the agent's environment.

For twenty years PowerShell was how we turned intent into reliable automation. Now PowerShell is how we *teach agents* to turn intent into reliable automation. Same skills. New surface.

PowerShell didn't get replaced.
It became the harness for the work
we already do.

What's next, not in scope today:

ADE — Agent Development Environment. Codex App for macOS/Windows. GitHub Copilot App (technical preview). It's not an IDE. It's where the agent lives.



Resources & Links

github.com/janegilring/presentations

Slides, demo scripts, and notes — published after the talk.

MCP & PowerShell

- [daxian-dbw/MCPServerPS](#)
- [dfinke/PSMCP](#)
- [KevinMarquette/pwsh.mcp.sdk](#)

Skills & agent mode

- [awesome-copilot.github.com](#)
- VS Code Agent Mode — Sessions 1–8
- Prompt engineering with Copilot (MS Learn)

Sources behind the quotes

- Snover & Finke — *Blown Away By AI*
- *Monad to Millions* panel
- Adam Jacob — *AI-Native Infrastructure Automation*
- Hanselman & Russinovich — *Learn to Have Taste*
- Steinberger on *Hanselminutes*

Thank you

Don't write more prompts. Write better skills.

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